VISHWASWARUP RATH

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EDUCATION

Manipal University Jaipur **October 2024 – June 2028**

- B.Tech in Computer Science and Engineering with Specialization in AI/ML **CGPA: 8.95**

EXPERIENCE

Trainee | Centre for Artificial Intelligence and Robotics (CAIR), DRDO, Bengaluru **June 2026 – July 2026**

- **Architected** a RAG-based system to process and retrieve information across **5+ multi-modal data types** (reports, documents, news articles, images), streamlining intelligence analysis workflows.
- Designed **3 core pipeline components** — information extraction, semantic retrieval, and structured knowledge generation — to support end-to-end intelligence analysis.

TECHNICAL SKILLS

Languages: Python, C++, SQL, JavaScript, TypeScript
Machine Learning & AI: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, Ollama, NLP, Retrieval-Augmented Generation (RAG), Embeddings, Semantic Search, OpenCV, YOLOv8
Data Science: Pandas, NumPy, Matplotlib
Backend: FastAPI, Flask, Node.js, REST APIs, Async Processing
Frontend: React, Next.js, Tailwind CSS
Databases: PostgreSQL, MongoDB, SQLite, Supabase, ChromaDB (Vector Database)
Data Engineering: ETL Pipelines, Web Scraping, Scrapy, BeautifulSoup, Selenium
Cloud & DevOps: AWS, Docker, Git, GitHub, CI/CD
Software Engineering: System Design, Microservices, Scalable Architecture

FEATURED PROJECTS

HappenMUJ – Campus Event Discovery and Community Platform [GitHub] [Live Demo] **October 2026**

- **Launched** a full-stack event discovery platform for MUJ with a **FastAPI + async PyMongo** backend on a **MongoDB replica set** and a **React + TypeScript** frontend, supporting event browsing, personal calendars, community discussions, and club/admin workflows.
- Designed a **9-collection document model** with embedding, extended-reference, and subset patterns, **21 indexes** (compound, partial, multikey, weighted text, TTL), **$jsonSchema** validation, **aggregation pipelines** (`$facet`, `$lookup`, `$setWindowFields`) for weighted Top-10 and faceted filters, multi-document transactions, and **GridFS** for poster storage.

NoCodeML – No-Code Machine Learning Experimentation Platform [GitHub] [Live Demo] **October 2026**

- **Developed** a full-stack no-code ML platform with a **Next.js 16 + React 19 + TypeScript** frontend and **FastAPI** backend guiding users through an **8-step workflow** (profiling, preprocessing, feature engineering, train/test split, model selection, training, results), deployed via **Docker** and **Vercel**.
- **Constructed** a leakage-safe **scikit-learn pipeline** training and comparing up to **9 model types** (Logistic/Linear/Ridge/Lasso Regression, KNN, Decision Tree, Random Forest, Gradient Boosting, **XGBoost**, SVM) with grid/random search, **ReportLab** PDF reports, **Supabase** row-level security, and **170+ pytest** tests.

IntelliRAG – Incident-Centric RAG for Terrorism Analysis [GitHub] **June 2026**

- **Deployed** a multi-modal RAG platform ingesting **5 document/media types** (PDFs, news articles, scanned documents, images, reports) into structured intelligence records via a **6-stage pipeline** — OCR, extraction, entity/relationship mapping, semantic retrieval, layered summarization, and analyst querying.
- Leveraged Python, local LLMs, vector search, and SQLite to power scalable intelligence workflows supporting threat assessment and decision-making under offline, air-gapped constraints.

Stock Recommender System [GitHub] **August 2026**

- **Delivered** a full-stack personalized stock recommendation platform on a **three-tier architecture** (React frontend, FastAPI REST backend, PostgreSQL database) with modular authentication, recommendation, admin, and health-check services.
- Implemented **JWT-based authentication (HS256)** with bcrypt hashing, **SQLAlchemy 2.0 ORM** with **Alembic** migrations, and a **React 18 + Vite** frontend with Context-based auth state serving personalized recommendations with confidence scores via a protected `/recommendations` endpoint.

Biterate – Restaurant Review Sentiment Analyzer [GitHub] [Live Demo] **April 2026**

- **Designed** an end-to-end NLP pipeline with **emoji-aware preprocessing**, negation handling, and **TF-IDF vectorization** (unigram, bigram, trigram) to prepare restaurant reviews for sentiment classification.
- Trained and benchmarked **5 models** — Logistic Regression, Naive Bayes, Linear SVC, XGBoost, and a **Soft Voting Ensemble** — via **5-fold Stratified CV**, achieving up to **88% accuracy** and **0.87 F1-score**, deployed through an interactive **Flask** dashboard for real-time prediction and analytics.

Clubs and Co-Curricular

First Year Co-ordinator | MUJ ACM SIGAI Student Chapter **Nov 2024 – May 2025**

- Selected among 14 candidates from 600+ applicants to coordinate college-level technical competitions and workshops, while contributing to website management and hosting initiatives.

Core Technical Team | AWS Cloud Club – MUJ **Nov 2025 – May 2026**

- **Developed** and maintained the official AWS Cloud Club MUJ website while contributing to the planning and execution of technical workshops, speaker sessions, and community events.

ACHIEVEMENTS AND CERTIFICATIONS

- Awarded the **Dean's List** for outstanding academic performance by securing the highest GPA across **two consecutive semesters** (1st and 2nd).
- Selected for **Amazon ML Summer School 2026**, a highly competitive machine learning program, among **3,000 participants** from over **134,000 applicants**.
- Successfully completed the **Machine Learning Specialization** by **Stanford University** and **DeepLearning.AI**, covering supervised learning, advanced learning algorithms, and unsupervised learning.
- Successfully completed the **Data Structures and Algorithm Design** course offered by **NPTEL**, earning the **ELITE** certification.
- Secured **2nd Place** at the **Innoverse Hackathon** by developing an innovative solution in a competitive team environment.